

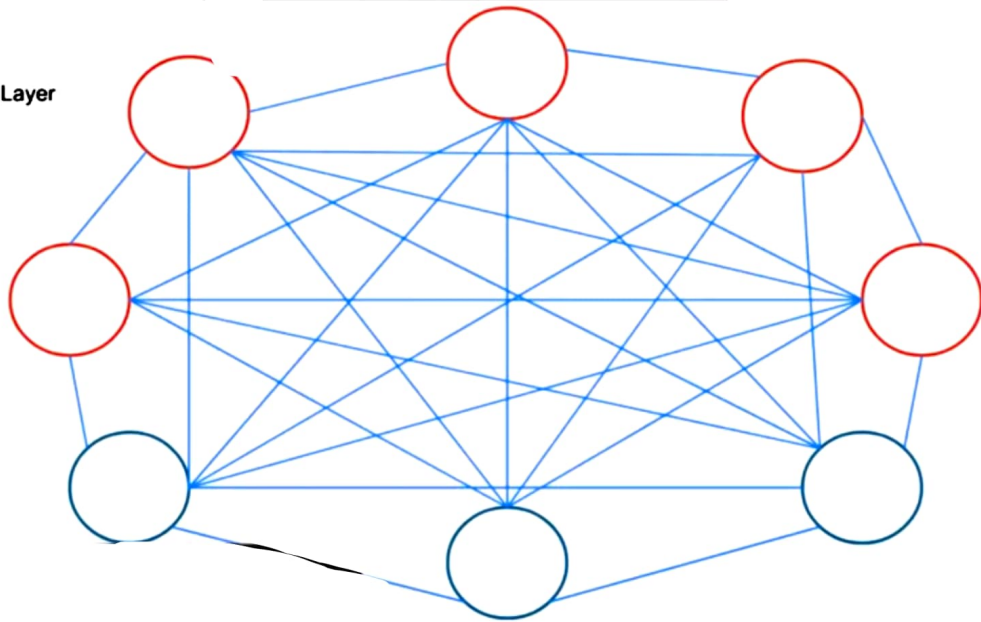
BOLTZMANN MACHINES

Boltzmann Machines

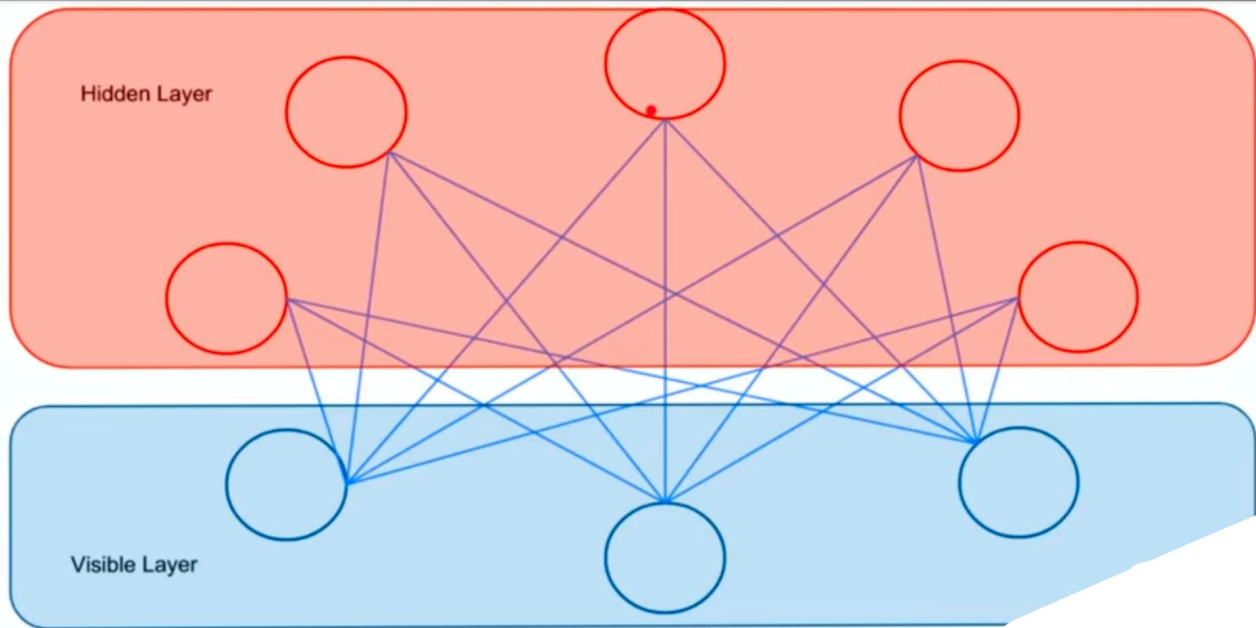
- A Boltzmann machine is a network of symmetrically connected, neuron-like units that make stochastic decisions about whether to be on or off
- Undirected models, where connection goes both ways
- Non-deterministic model
- Purpose is to optimize the solution
- Discover features from datasets composed of binary vectors

Boltzmann Machine

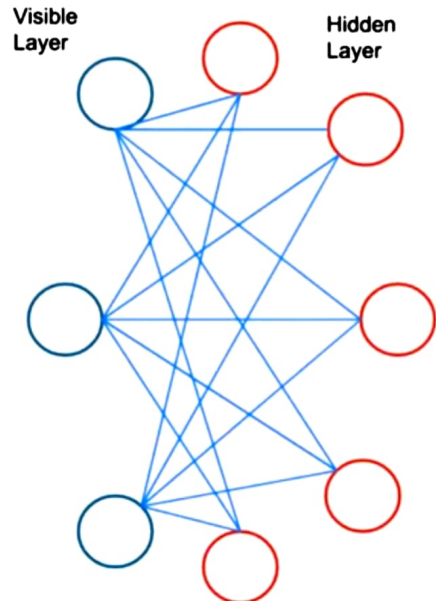
Hidden Layer



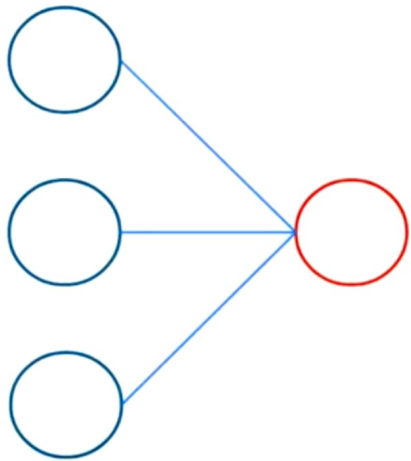
Restricted Boltzmann Machine — RBM



Restricted Boltzmann Machine — RBM (Continued)

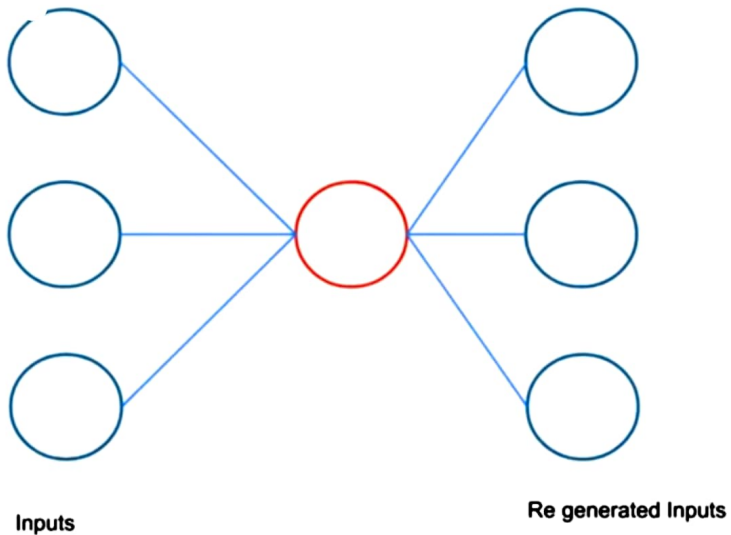


Understanding How Boltzmann Machine Learns

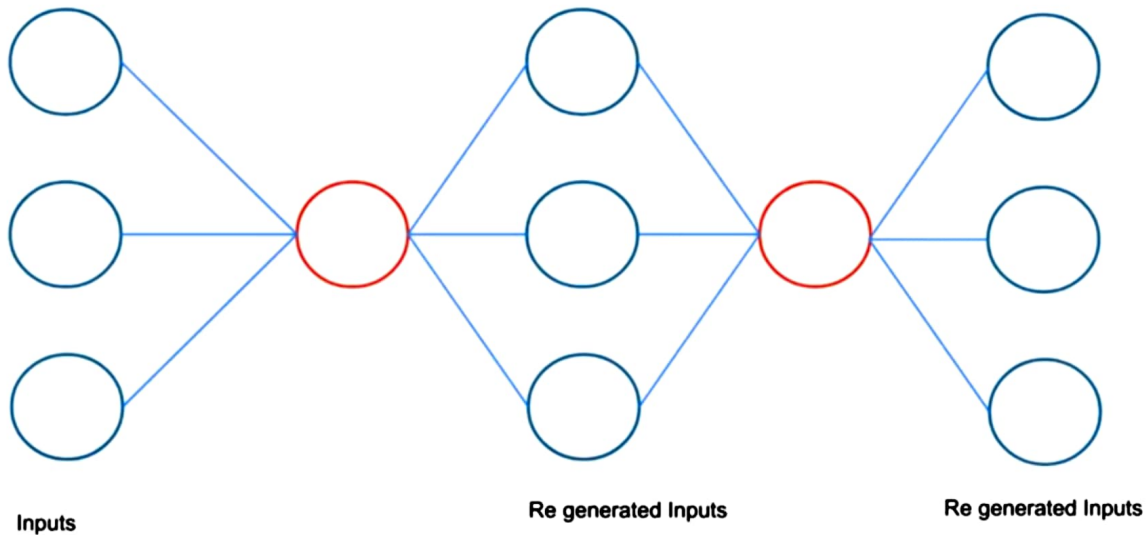


Inputs

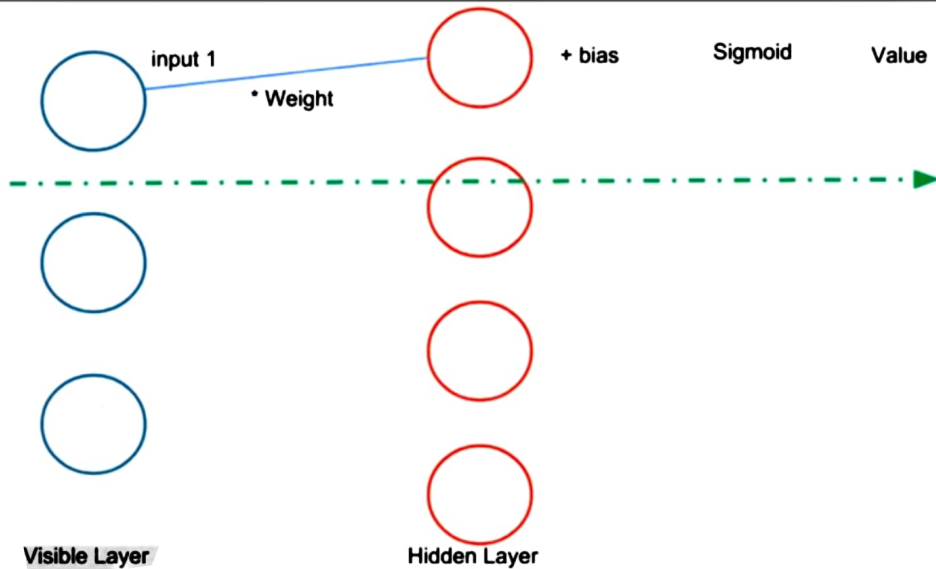
Understanding How Boltzmann Machine Learns



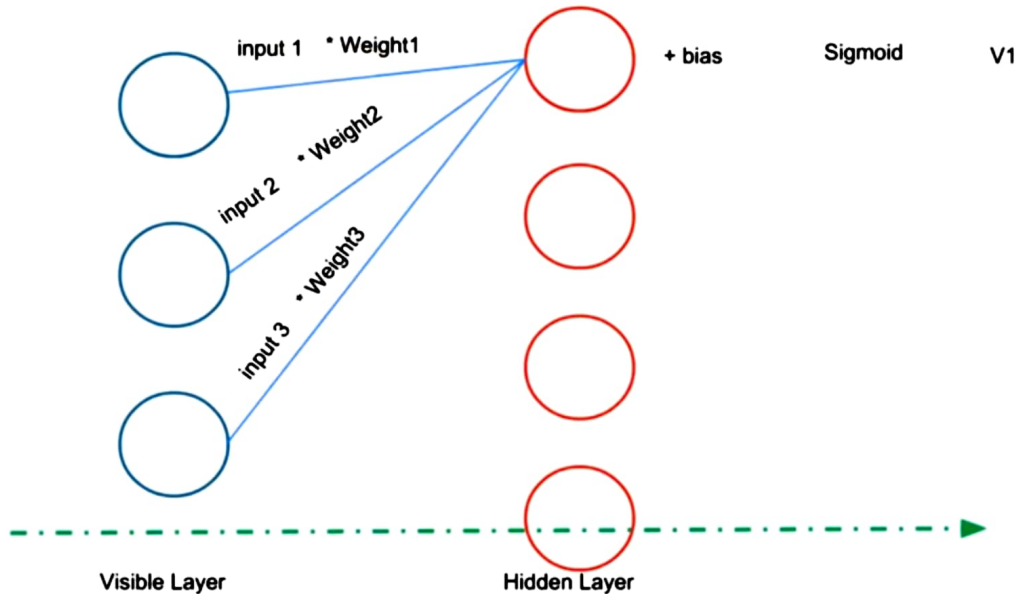
Understanding How Boltzmann Machine Learns



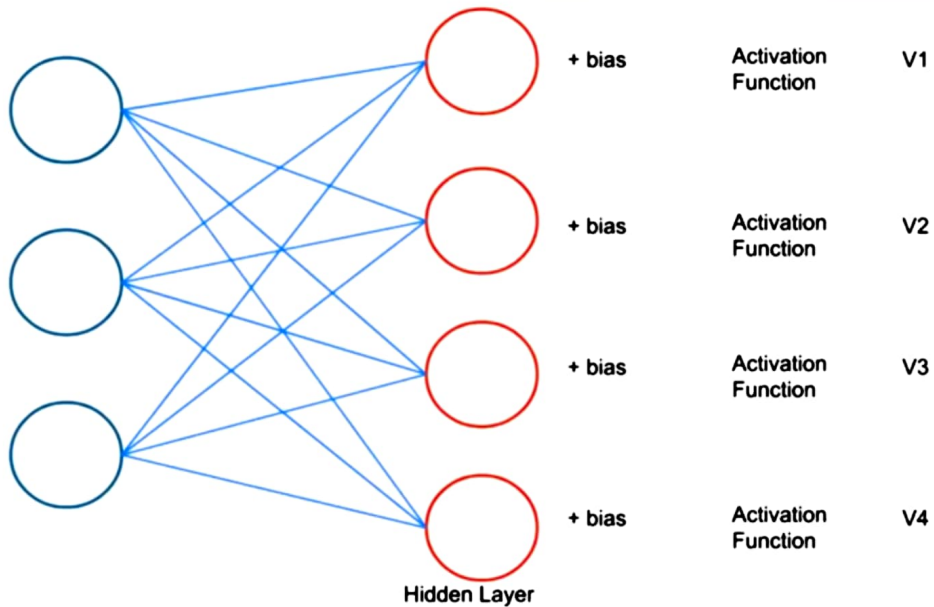
Understanding How Boltzmann Machine Learns



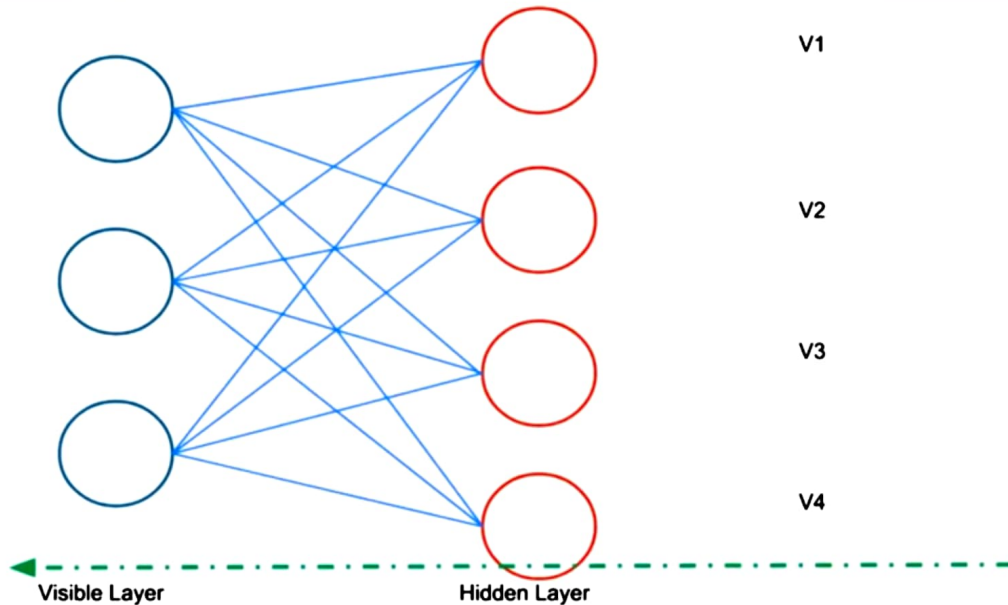
Understanding How Boltzmann Machine Learns



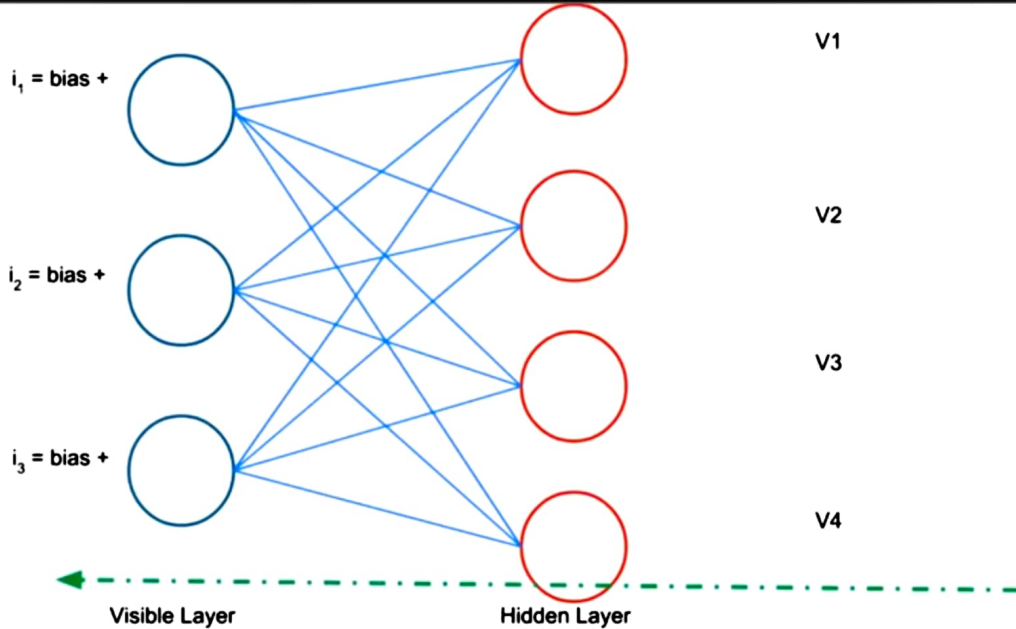
Understanding How Boltzmann Machine Learns



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Understanding How Boltzmann Machine Learns



$$\sum_{\mathbf{v} \in \text{data}} \frac{\partial \log P(\mathbf{v})}{\partial w_{ij}} = \langle s_i s_j \rangle_{\text{data}} - \langle s_i s_j \rangle_{\text{model}}$$